CT029-3-2

HAND OUT DATE:

HAND IN DATE:

WEIGHTAGE:

INSTRUCTIONS TO CANDIDATES:

1.Submit your assignment online in Moodle unless advised otherwise
2. Students are advised to underpin their answers with the use of references (cited using the 7th edition APA (American Psychological Association) Referencing style)
3. Late submission will be awarded zero (0) unless Extenuating Circumstances (EC) are upheld.
4.Cases of plagiarism will be penalized
5.You must obtain 50% overall to pass this module.

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INTRODUCTIUON 1.0

Python offers powerful tools for image editing, making it an essential skill for artists and content creators. The Pillow library allows users to easily open, manipulate, and save images in various formats. For more advanced techniques, libraries like OpenCV and Matplotlib provide additional capabilities, including computer vision and data visualization. In this guide, you'll learn how to use Python to edit images and apply effects, from resizing and cropping to adding filters and overlays. Whether you're a beginner or experienced programmer, you'll discover how to transform your photos with just a few lines of code therefore able to show the story. In a war-torn world, Eli navigates the desolate desert, seeking hope in the ruins of Haven ridge. Amidst radiation and decay, he finds a community of survivors led by Mira. Together, they confront dangers, forge bonds, and dream of rebuilding a future where hope can flourish amidst the ashes of despair.

STORY LINE 2.0

In a world ravaged by war, where nations have turned against each other and the skies are perpetually choked with ash, Eli finds himself living in a dilapidated shack in the heart of a vast desert. Once a thriving community, the area has been reduced to desolation, its remnants scattered like memories in the wind. The war, fuelled by greed and power, has unleashed a catastrophic wave of destruction, resulting in an apocalypse that has left the land barren and the people scattered.

Eli, a former engineer, has become a reluctant survivor, navigating the harsh realities of a world where resources are scarce, and danger lurks at every turn. His shack, a makeshift home adorned with scavenged materials, offers little comfort but serves as a sanctuary from the chaos outside. The only company he has is a small radio that occasionally crackles to life with static-laden broadcasts, remnants of a time long gone.

One fateful morning, Eli receives a faint transmission hinting at a nearby town, Haven ridge, which has been affected by radiation but may still hold valuable supplies and remnants of civilization. Driven by desperation and the hope of finding something—anything—to improve his dire situation, Eli sets out on foot along the abandoned railway that once connected the towns of the region.

As he walks along the rusting tracks, Eli reflects on the world that was. The railway, once a lifeline for communities, now stands as a ghostly reminder of humanity's hubris. Along the way, he encounters remnants of the past: derelict trains, overgrown with sand and weeds, and faded posters advertising a brighter future that now feels like a cruel joke.

As Eli approaches Haven ridge, he notices signs of decay—buildings crumbling under the weight of time, the air thick with an unsettling silence. The town, shrouded in an eerie glow from the lingering radiation, is a haunting sight. He must navigate the dangers of radiation exposure while searching for food, water, and any sign of life.

In Haven ridge, Eli encounters a small group of survivors who have banded together, each carrying their own scars from the war and the apocalypse. Among them is Mira, a fiercely intelligent woman with a knack for scavenging and a deep-seated hope for the future. As Eli integrates into the group, he learns of their struggles and the harsh realities of living in a radiation-affected zone. Together, they devise a plan to fortify their makeshift community and create a sustainable way of life.

However, their fragile existence is threatened by a ruthless band of raiders who roam the desert, preying on the weak and desperate. As tensions rise, Eli must confront his past and the choices that led him to this moment. In a climactic showdown, he and the survivors band together to defend their newfound home, forging bonds of trust and resilience in the face of overwhelming odds.

In the aftermath of the conflict, Eli realizes that survival is not just about enduring but about rebuilding and finding purpose in a shattered world. With Mira by his side, they begin to dream of a future beyond mere survival—a future where hope can bloom even in the harshest of landscapes.

As the sun sets over the desert, casting long shadows over the ruins of Haven ridge, Eli stands on the threshold of a new beginning, determined to create a life worth living in the echoes of the past.

IMAGE PROCESSING & SPECIAL EFFECT TECHNIQUES DISCUSSION 3.0

DARREN HON HOW SHENG 3.1

The picture shows a mushroom cloud rising above a cityscape. The cloud is large and powerful, obscuring much of the city in smoke and debris. This image is a stark reminder of the destructive power of nuclear weapons. It is a powerful image that evokes feelings of fear, anxiety, and sorrow. It is important to remember the human cost of war, and to strive for a world free of such destructive weapons.

Resizing

The background image is resized to match the dimensions of the foreground image. This ensures that both images align correctly when they are combined.

Darkening the Background

The script reduces the brightness of the background image by multiplying its pixel values by a darkening factor (0.5 in this case). This creates a more dramatic effect when the foreground (the explosion) is added on top.

Green Screen Removal

The script defines a colour range for green (the typical colour of a green screen) using RGB values. It uses OpenCV's cv2.inRange() function to create a binary mask that identifies pixels in the foreground image that fall within the specified green colour range.

The inverse of this mask is created using cv2.bitwise_not (), allowing for the selection of pixels that are not green.

Mask Application

The masks created are applied to both the foreground and background images:

The foreground mask (masking) is used to retain only the non-green pixels of the foreground image.

The background mask is used to retain the green pixels from the background image, effectively allowing the foreground to overlay on the background.

Image Combination

The cleaned foreground and background images are combined using cv2.add (), which performs a pixel-wise addition of the two images. This results in a single composite image where the explosion appears to be part of the city scene.

This image shows a mountainside. The canyon is deep and narrow, with steep, rocky walls. The surrounding landscape is rugged and wild, with grassy hills and patches of snow. The sky is a clear blue, and the sun is shining brightly. The image captures the raw beauty of nature, and it is a testament to the power of the elements. After when the bomb hits the middle of the city it causes the beautiful mountain side to shake violently and causing the crack to widen.

Zoom Effect:

Cropping: The image is cropped to create a zoom effect. The script calculates the new dimensions based on a zoom factor (1.2), which means the image will be zoomed in by 20%. The cropping coordinates are calculated to centre the zoom effect.

Resizing: After cropping, the image is resized back to its original dimensions using the resize () method with the Image.LANCZOS filter, which provides high-quality down sampling.

Shaking Effect:

Frame Creation: The script creates a series of frames to simulate a shaking effect. It generates 20 frames in total.

Random Shifting: For each frame, the script randomly shifts the zoomed image both horizontally and vertically within a defined range (shake range). The np. random. radiant () function is used to generate random integers for the shifts.

Pasting Shifted Images: A new blank image (shifted_image) is created for each frame, and the zoomed image is pasted onto this new image at the randomly determined shifted position.

GIF Creation:

The frames are compiled into a GIF using the save () method. The save-all=True argument ensures that all frames are saved in the GIF. The append images argument adds the subsequent frames to the first one.

The duration parameter controls the time each frame is displayed (30 milliseconds in this case), and loop=0 makes the GIF loop indefinitely.

YEE ZHENG XUN 3.2

A vintage car, rusted and worn, drives down a deserted street in the heart of a city. The buildings on either side, tall and stately, are seemingly abandoned, their windows dark and empty. The air hangs heavy, filled with a sense of decay and forgotten history. The scene evokes a sense of mystery and intrigue, prompting the viewer to wonder what events led to this desolate state.

Yellow Filter: The image has a yellow filter added, making the photo reminiscent of a post-apocalyptic city.

Cropped car: The car’s original picture was first applied with a Bilateral Filter to smoothen it out. Then it’s background was removed, and it was then added into the photo using code.

Smoke: The smoke’s background was removed, after which a transparent effect was added by changing the alpha value. Then it was added to the photo.

This picture is a picture of the survivor trying to fight the cure to the zombie virus. This lab that still contains all the equipment that it has is a little bit dilapidated. Therefore in the picture you can see it is dark because electricity can't be used in the lab and the only thing that is illuminating in the lab is the vile for the cure.

Laboratory: The laboratory was darkened by scaling down the alpha of the image.

Scientist: Scientist is cropped and added to image. The beaker’s glow was added using gaussian blur.

YAP ZHENG FUNG 3.3

The picture shows a city street engulfed in flames and smoke. A group of zombies are walking down the middle of the street. Buildings are burning in the background. The scene is chaotic and apocalyptic. Then the zombies notice the scavenger named Eli.

Image Resizing

The zombie images are resized using a scale factor (0.5) to ensure they fit appropriately within the scene. The Image.LANCZOS filter is used for high-quality resizing.

Background Removal

A function called remove background is defined to eliminate white and light gray backgrounds from the images. This is achieved by creating a transparency mask based on a colour threshold (200), which identifies pixels that are predominantly white or light Gray. These pixels are then made fully transparent.

Colour Adjustment

The adjust_color function is used to modify the brightness, contrast, and colour saturation of the zombie images. This helps the zombies blend better with the environment. The Image Enhance module is used to adjust these properties:

Brightness: Reduced to 80% of the original.

Contrast: Increased to 120% of the original.

Colour Saturation: Reduced to 80% of the original.

Background Darkening

The city image is darkened using the darken background function to simulate an overcast or rainy effect. This is accomplished by reducing the brightness of the city image to 65% of its original value.

Fire Image Darkening

The fire image is also darkened using the darken fire function, which reduces its brightness to 70% of the original to make it less intense and more realistic within the scene.

Positioning Elements

The script calculates positions for the zombies based on the dimensions of the city image. The zombies are positioned towards the bottom of the image, slightly offset from the centre to create a more dynamic composition.

8. Smoke Resizing for Depth Effect

The smoke image is resized to create a sense of depth in the scene. Two different scales are used: a larger scale for the smoke that appears closer to the viewer and a smaller scale for the smoke that appears further away.

9. Static Smoke Layers

The smoke images are positioned in the scene to create the illusion of depth. The left smoke is smaller and positioned further back, while the right smoke is larger and positioned closer to the viewer.

10. Fire Image Resizing and Positioning

The fire image is resized to stretch across the width of the city image and positioned at a height that allows it to blend well with the other elements.

The image shows a man in a white hazmat suit running away from a burning car. The car is engulfed in flames and smoke, and the man is looking over his shoulder as he runs. The scene is set against a backdrop of a snowy, industrial area. The man is likely running away from a dangerous situation, such as a zombie outbreak or an explosion. The image is filled with a sense of urgency and chaos.

Image Composition

A new image (final image) is created by copying the city image. The fire image is pasted on top of the city image first, followed by the smoke layers and finally the zombie images. The use of transparency in the pasted images allows for a seamless integration of all elements.

The image shows a man in a white hazmat suit running away from a burning car. The car is engulfed in flames and smoke, and the man is looking over his shoulder as he runs. The scene is set against a backdrop of a snowy, industry

Background Darkening:

The background image is darkened by applying a semi-transparent black overlay. This is done by creating a new RGBA image filled with black and an alpha value of 100 (semi-transparent) and pasting it over the original background.

Car Image Processing:

The car image is converted to RGBA mode to support transparency. A threshold is applied to identify and remove white pixels (making them transparent). This is done by iterating over the pixel data and replacing white pixels with transparent pixels.

The car image is then resized to be five times larger to make it a more prominent feature in the scene.

Fire Image Processing:

The fire image is resized to 120% of the car's height and converted to RGBA mode. The alpha channel is adjusted to reduce the opacity to 90% to blend better with the background.

Smoke Image Processing:

Like the fire image, the smoke image is processed to remove its white background using a threshold. The smoke image is resized to be twice its height and its opacity is reduced to 50% to create a more realistic effect.

Hazmat Suit Image Processing:

The hazmat suit image is converted to RGBA mode, and a threshold is applied to remove green pixels (making them transparent). This is useful for images where the subject is wearing a green suit.

The hazmat image is resized to increase its dimensions by 1200 pixels in both width and height, making it significantly larger.

Positioning Elements:

The script calculates the position for the car at the bottom-right of the background and the hazmat figure on the left side. This ensures that elements are placed appropriately within the scene.

Pasting Elements onto the Background:

The processed car, fire, smoke, and hazmat images are pasted onto the darkened background at their respective positions. The paste method of the PIL library allows for the use of transparency in the pasted images.

Creating an Animated GIF with Snow Effect:

The script creates an animated GIF that simulates falling snow. It initializes a list of snowflakes with random positions and sets a falling speed.

For each frame of the animation:

A copy of the background is created.

The hazmat figure is pasted onto the frame.

The positions of the snowflakes are updated, and new snowflake positions are calculated. If a snowflake moves beyond the bottom of the background, it is reset to a random position at the top.

Snowflakes are drawn as white circles on the frame using the draw. ellipse method.

The frames are stored in a list for later use in creating the GIF.

Memory Management:

The script includes memory management techniques such as deleting the draw object and invoking garbage collection (gc. collect ()) to free up memory resources during the frame generation process.

Saving the Animated GIF:

Finally, the frames are saved as an animated GIF using the save method of the PIL library. The GIF is configured to have a total duration of 20 seconds, with each frame displayed for approximately 333 milliseconds.

Yehia Rasheed Yehia Al-haddad 3.4

This picture shows the survivor walking back to his house. By following the train tracks and walking for many miles. He is forced to wear a raincoat and carry many heavy gears in this picture the camera is also affected by radiation therefore the picture is blur.

A bilateral filter is a non-linear image processing technique that smooths images while preserving edges. It combines spatial proximity and intensity similarity to assess the influence of neighboring pixels on the filtered pixel, effectively reducing noise while maintaining important features. This makes it particularly useful in applications such as computer vision and image denoising.

When the survivor has finally made it to his shack in the middle of the desert this is the only location that has not been affected by radiation before the outbreak of the zombie virus. At this location the survivor can finally breathe a sigh of relief and having another spare car he is able to go at explore once again.

Cropped car: The car’s original picture was first applied with a Bilateral Filter to smoothen it out. Then it’s background was removed, and it was then added into the photo using code.

Clouds: The clouds first had a bilateral filter added, then it’s black background was removed. It was finally added to the photo.

Man: The man first added a bilateral filter, then it was cropped, it's background was removed and it was added to the photo.

Particles: A set number of particles are set, the particles will be loaded from the top of the screen and when they touch the edge of the picture more will be loaded at the top of the screen.

CONCLUSION 4.0

In this project, we effectively explored and implemented various image editing techniques using Python, leveraging libraries such as Pillow and OpenCV. We showcased Python's versatility in handling a range of image processing tasks, including resizing, filtering, cropping, and colour adjustments. The project highlighted not only the technical capabilities of these libraries but also the ease of automating and customizing image editing workflows for practical applications. By the end, we gained a deeper understanding of image manipulation principles and recognized how Python's extensive library ecosystem can be utilized to create powerful tools for both simple and advanced editing tasks. Looking forward, to have opportunities to enhance our work by integrating machine learning models for automatic image enhancement or analysis, optimizing processing speeds for large image sets, and exploring additional libraries like scikit-image. We could also consider developing user interfaces for more interactive applications. Given the increasing demand for visual content across various industries, the ability to efficiently edit images programmatically is a valuable skill. This project has laid a solid foundation for further exploration and development in the field of image editing.

REFRENCES 6.0

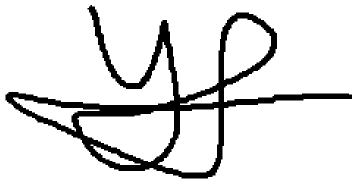
Nuke Image	https://www.google.com/search?sca_esv=f844996498a82a61&sxsrf=ADLYWIJPLiG4-1lD1cFKjGkiwQVM_qIORA:1731926813684&q=nuke+greenscreen&udm=2&fbs=AEQNm0Aa4sjWe7Rqy32pFwRj0UkWwAFG7ranuZ26H8lR7pf_8A-INd5ojJqZL-l5U4A6ut-tz5q4L-0jabWW_E10_lCRullqVtI43dS5HsVCwZGXLSXuFc6dLOZF8badrZ5cP8rL_H--gLY81z-hX29RWDBOp90gRP1Y-qZx21uE6qiEo4HRhHtZDyrQXppD-XuSPdExn-FnnIr8XckHnqlfxVr5zQH0Hg&sa=X&ved=2ahUKEwiuwIaU2uWJAxVrSGwGHRi6MoUQtKgLegQIFxAB&biw=1920&bih=953&dpr=1#imgrc=9JVXmJxgC6s1xM&imgdii=Zfv7rsOEtSDp8M
City Image	https://www.pexels.com/photo/empire-state-building-new-york-466685/

Fire Image	https://www.myxxgirl.com/realistic-fire-gif-png/sQuiRtcHM6Ly9pbWFnZS5wbmdhYWEuY29tLzgwLzM0MzA4MC1taWRkbGUucG5nfulL-hDgiffulL-hDsQuiRtcHM6Ly90c2UzLm1tLmJpbmcubmV0L3RoP2lkPU9JUC5hcmpzaUZ4MXpEWUxxQcUhvNWtWeHlBSGFGViZwaWQ9MilFUuMQ.gif.html
Fire Spreading Image	https://stock.adobe.com/search?k=fire+transparent&asset_id=723052277
Smoke Image	https://www.pinterest.com/pin/pinterest--861876447431463970/
Earthsplitting apart Image	https://www.dreamstime.com/breathtaking-summer-landscape-mountains-covered-green-icelandic-moss-deep-ravine-ground-highlands-iceland-image130547487
City Image	https://www.gettyimages.com/videos/new-york-street
Zombie Image	https://stock.adobe.com/search?k=zombies%20crowd
Factory Image	https://www.pexels.com/photo/abandoned-warehouse-facade-12069529/
Dude in Hazmat Image	https://www.shutterstock.com/video/clip-1051575136-medical-doctor-hazmat-clothes-pointing-on-imaginary

Broken Image	https://pngtree.com/freepng/car-crash-drunk-impact_13041340.html
City	https://www.google.com/search?sca_esv=6e20b04bcc434b58&q=empty+street&udm=2&fbs=AEQNm0Aa4sjWe7Rqy32pFwRj0UkWd8nbOJfsBGGB5IQQO6L3JyJJclJuzBPl12qJyPx7ESI2ylY26TJcHuYtL_AqwMrMI9hiawHArZ71iNhwYF-kuspHt-TkeEPl4JkkajwieOI4lG9u3vN4AXJImJx4HjSS72bltSSs9PVr-DnhkvVyxK55OA2zERKRHIpADnPNKHouRKcAv2X4x_ZMgKmZ7bP5US7FyQ&sa=X&sqi=2&ved=2ahUKEwjajqesl9yJAxXER2wGHa2QI3oQtKgLegQIExAB&biw=1707&bih=932&dpr=1.5#vhid=RXwSM2b1oO-QVM&vssid=mosaic
Rusty Car	https://www.istockphoto.com/photo/old-rusty-classic-car-isolated-on-white-background-old-rusty-ancient-car-isolated-gm974657464-265145687?searchscope=image%2Cfilm
Rusty Car 2	https://stock.adobe.com/search?k=rusty+cars
Cloud	https://www.vecteezy.com/png/26774603-cloud-png-with-ai-generated
Smoke	https://pngtree.com/freepng/realistic-smoke-effect_14670417.html
Man	https://www.google.com/search?q=post-apocalyptic+man+backshot&sca_esv=ff4380769abef046&udm=2&biw=1707&bih=932&ei=zIk4Z_DPGb-YseMP26um0QY&ved=0ahUKEwjw5Y2l5-CJAxU_TGwGHduVKWoQ4dUDCBA&uact=5&oq=post-apocalyptic+man+backshot&gs_lp=EgNpbWciHXBvc3QtYXBvY2FseXB0aWMgbWFuIGJhY2tzaG90SOcRUKwEWNoQcAF4AJABAJgBKqABvwKqAQE5uAEDyAEA-AEBmAIDoAJQwgIFEAAYgATCAgQQABgewgIGEAAYChgewgIGEAAYCBgemAMAiAYBkgcBM6AH_ws&sclient=img#vhid=fyTL4EFrzwA7rM&vssid=mosaic
Shack	https://stablediffusionweb.com/it/image/7780623-solitary-desert-shack

Laboratory	https://www.gettyimages.co.uk/detail/photo/modern-empty-science-laboratory-with-microscopes-royalty-free-image/1951546291?adppopup=true
Scientist	https://www.gettyimages.com/detail/photo/japanese-scientist-working-in-laboratory-royalty-free-image/463245553?adppopup=true
Radiation	https://wallpaperaccess.com/full/2808834.jpg
Highway	https://wallpaperaccess.com/full/2020619.jpg
Bridge	https://wallpaperaccess.com/full/2020714.jpg
Clouds	https://wallpaperaccess.com/full/4389461.jpg
Abandoned car	https://wallpaperaccess.com/full/6463129.jpg
woman	https://wallpaperaccess.com/full/1810890.jpg
Destroyed city	https://wallpaperaccess.com/full/1373364.jpg

WORKLOAD MATRIX 7.0

NAME	TPNUMBER	CONTRIBUTION	SIGNATURE
YEE ZHENG XUN	**TP069088**	Story Chapter 2& 4 Workload Matrix & Appendices Image processing Technique	
DARREN HON HOE SHENG	**TP069062**	Story Chapter 1 Introduction Image Processing Technique	
YAP ZHEN FUNG	**TP066371**	Story Chapter 3 & 4 Compiling Group Programs Image Processing Technique	
YEHIA RASHEED YEHIA AL HADDAD	**TP072510**	Story Chapter 5 Conclusion Image Processing Technique